In [109…

```python
import pandas as pd
import numpy as np
```

4)

In [110…

```python
df = pd.read_excel("inputs/Domestic_shares.xlsx", header=2)
df = df.drop(columns=['Unnamed: 0'])
df = df.set_index('Country')
```

In [111…

```python
df = df[df.index.isin(["USA", "CAN", "MEX", "CHN"])][[2000, 2014]]
```

In [112…

```python
countries = ['USA', 'CAN', 'MEX', 'CHN']
years = [2000, 2014]
epsilons = [4, 5]

rows = []

for c in countries:
    row = {'Country': c}

    for y in years:
        lam = df.loc[c, y]
        row[f'lambda_{y}'] = lam

        for e in epsilons:
            gft = (lam**(-1/e) - 1) * 100
            row[f'GFT_{y}_e{e}'] = gft

    rows.append(row)

out = pd.DataFrame(rows)

out = out.rename(columns={
    'lambda_2000': 'λii (2000)',
    'GFT_2000_e4': 'GFT (2000) ε=4',
    'GFT_2000_e5': 'GFT (2000) ε=5',
    'lambda_2014': 'λii (2014)',
    'GFT_2014_e4': 'GFT (2014) ε=4',
    'GFT_2014_e5': 'GFT (2014) ε=5',
})

out_display = out.copy()

for col in out_display.columns[1:]:
    out_display[col] = out_display[col].round(3)

out_display
```

Out[112…

	Country	λii (2000)	GFT (2000) ε=4	GFT (2000) ε=5	λii (2014)	GFT (2014) ε=4	GFT (2014) ε=5
0	USA	0.932	1.775	1.418	0.923	2.011	1.606
1	CAN	0.816	5.228	4.161	0.832	4.719	3.758
2	MEX	0.845	4.310	3.433	0.830	4.775	3.802
3	CHN	0.931	1.795	1.433	0.941	1.536	1.227

6)

In [113…

```
df = pd.read_stata("inputs/WIOT2014_October16_ROW.dta")

df = df[df['Country']!='TOT']

df = df.groupby("Country").sum().reset_index()

for country in list(df['Country']):
    df[country] = df[[col for col in df.columns if col.startswith("v" + country)]].

df = df[["Country"] + list(df['Country'])]
```

In [114…

```
df2 = df.iloc[:,~(df.columns.isin(['USA','CHN','CAN','MEX']))]
df3 = df2['Country'].copy().to_frame()
df['ROW'] = np.sum(df2.iloc[:,1:],axis = 1)
```

In [115…

```
exclude = ["USA", "CHN", "CAN", "MEX"]

subset = df.loc[~df["Country"].isin(exclude)]

new_row = subset.sum(axis=0)

agg_df = new_row.to_frame().T

agg_df = agg_df[df.columns]

country_col = df.columns[0]
agg_df[country_col] = "AGG_OTHERS"
```

In [116…

```
us_2014 = {"USA": [
    df[df["Country"] == "USA"]["USA"].values[0] / df["USA"].sum(),
    df[df["Country"] == "CHN"]["USA"].values[0] / df["USA"].sum(),
    df[df["Country"] == "MEX"]["USA"].values[0] / df["USA"].sum(),
    df[df["Country"] == "CAN"]["USA"].values[0] / df["USA"].sum(),
    agg_df[agg_df["Country"] == "AGG_OTHERS"]["USA"].values[0] / df["USA"].sum()
],
     "CHN": [
    df[df["Country"] == "USA"]["CHN"].values[0] / df["CHN"].sum(),
    df[df["Country"] == "CHN"]["CHN"].values[0] / df["CHN"].sum(),
    df[df["Country"] == "MEX"]["CHN"].values[0] / df["CHN"].sum(),
```

```
    df[df["Country"] == "CAN"]["CHN"].values[0] / df["CHN"].sum(),
    agg_df[agg_df["Country"] == "AGG_OTHERS"]["CHN"].values[0] / df["CHN"].sum()
],
     "MEX": [
    df[df["Country"] == "USA"]["MEX"].values[0] / df["MEX"].sum(),
    df[df["Country"] == "CHN"]["MEX"].values[0] / df["MEX"].sum(),
    df[df["Country"] == "MEX"]["MEX"].values[0] / df["MEX"].sum(),
    df[df["Country"] == "CAN"]["MEX"].values[0] / df["MEX"].sum(),
    agg_df[agg_df["Country"] == "AGG_OTHERS"]["MEX"].values[0] / df["MEX"].sum()
],
     "CAN": [
    df[df["Country"] == "USA"]["CAN"].values[0] / df["CAN"].sum(),
    df[df["Country"] == "CHN"]["CAN"].values[0] / df["CAN"].sum(),
    df[df["Country"] == "MEX"]["CAN"].values[0] / df["CAN"].sum(),
    df[df["Country"] == "CAN"]["CAN"].values[0] / df["CAN"].sum(),
    agg_df[agg_df["Country"] == "AGG_OTHERS"]["CAN"].values[0] / df["CAN"].sum()
]}

us_2000 = np.array([[0.932,0.003,0.007,0.012,0.046],
                [0.004,0.931,0.0000001,0.001,0.064],
                [0.097,0.002,0.845,0.003,0.054],
                [0.114,0.004,0.005,0.816,0.061]])

df_l2000 = pd.DataFrame(us_2000, index=["USA", "CHN", "MEX", "CAN"], columns=["USA"

df_l2014 = pd.DataFrame(us_2014, index=["USA", "CHN", "MEX", "CAN", "ROW"]).T
```

In [117…

```
## Expenditure shares 2000

df_l2000
```

Out[117…

	USA	CHN	MEX	CAN	ROW
USA	0.932	0.003	7.000000e-03	0.012	0.046
CHN	0.004	0.931	1.000000e-07	0.001	0.064
MEX	0.097	0.002	8.450000e-01	0.003	0.054
CAN	0.114	0.004	5.000000e-03	0.816	0.061

In [121…

```
## Expenditure shares 2014

df_l2014.round(3)
```

Out[121…

	USA	CHN	MEX	CAN	ROW
USA	0.923	0.011	0.009	0.011	0.046
CHN	0.004	0.941	0.000	0.001	0.055
MEX	0.084	0.018	0.830	0.004	0.064
CAN	0.090	0.015	0.006	0.832	0.057

7)

$$\frac{1}{\varepsilon}\sum_{n\neq i}\left(\frac{\lambda_{in,t}}{\lambda_{ii,t}}\right)\frac{\lambda_{in,t+1}-\lambda_{in,t}}{\lambda_{in,t}}$$

In [122…

```
## Individual contributions

e = 5.0

l2014 = df_l2014.to_numpy()

l2000 = df_l2000.to_numpy()

t7 = np.full(shape=(4,5),fill_value=np.nan)

for i in range(0,4):
    for j in range(0,5):
        arr = np.concatenate([np.arange(0, i), np.arange(i + 1, 5)])
        t7[i,j] = 1/e*l2000[i,j]/l2000[i,i]*(l2014[i,j]-l2000[i,j])/l2000[i,j]

df7 = pd.DataFrame(t7,columns=['US Cont','CHN Cont','MEX Cont','CAN Cont','ROW Cont

df7 = df7[["US Cont", "CAN Cont", "MEX Cont", "CHN Cont", "ROW Cont"]]

df7 = df7.reindex(["USA", "CAN", "MEX", "CHN"])

df7.round(4)
```

Out[122…

	US Cont	CAN Cont	MEX Cont	CHN Cont	ROW Cont
USA	-0.0018	-0.0002	0.0003	0.0017	-0.0000
CAN	-0.0058	0.0038	0.0003	0.0028	-0.0011
MEX	-0.0031	0.0002	-0.0036	0.0038	0.0024
CHN	-0.0001	-0.0001	0.0001	0.0021	-0.0020

In [123…

```
## GFT and contributions as ratio

e = 5.0

l2014 = df_l2014.to_numpy()

l2000 = df_l2000.to_numpy()

t7 = np.full(shape=(4,6),fill_value=np.nan)

for i in range(0,4):
    arr = np.concatenate([np.arange(0, i), np.arange(i + 1, 5)])
```

```
    t7[i,0] = 1/e*np.sum(l2000[i,arr]/l2000[i,i]*(l2014[i,arr]-l2000[i,arr])/l2000[

for i in range(0,4):
    for j in range(0,5):
            if j == i:
                 continue
            t7[i,j+1] = 1/e*l2000[i,j]/l2000[i,i]*(l2014[i,j]-l2000[i,j])/l2000[i,j

df7 = pd.DataFrame(t7,columns=['GFT','US Cont','CHN Cont','MEX Cont','CAN Cont','RO

df7 = df7[["GFT", "US Cont", "CAN Cont", "MEX Cont", "CHN Cont", "ROW Cont"]]

df7 = df7.reindex(["USA", "CAN", "MEX", "CHN"])

df7.round(4)
```

Out[123…

	GFT	US Cont	CAN Cont	MEX Cont	CHN Cont	ROW Cont
USA	0.0018	NaN	-0.1002	0.1755	0.9405	-0.0159
CAN	-0.0038	1.5243	NaN	-0.0714	-0.7297	0.2769
MEX	0.0034	-0.9095	0.0619	NaN	1.1306	0.7170
CHN	-0.0021	0.0410	0.0434	-0.0240	NaN	0.9396